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Release notes for community-compatible edition 26.3

Updated at: 2026-08-03 10:30:11

This topic describes the major changes in Alibaba Cloud ClickHouse community-compatible edition 26.3 LTS, including new features, performance optimizations, and improvements.

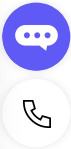
Overview

Alibaba Cloud ClickHouse community-compatible edition releases the new long-term support (LTS) version 26.3. As a major

On this page (1)

- Overview
- Key highlights
 - Data lake ecosystem: Apache Paimon and unified Catalog
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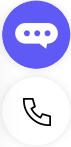
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annual release following 25.3 LTS, version 26.3 focuses on unified batch-streaming data lake, native inverted index, high-concurrency write stability, complex query optimization, and open ecosystem extensibility.

Key highlights

C a t e g o r y	Ke y im pr ov e m e n t s	Bu sin e s s v a l u e
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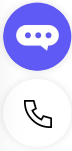
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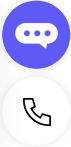

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O p e n e c o s y	Co m pa tib ilit y wi th 31	Mi gr at e cr os s- en gin

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**Data lake
ecosystem:
Apache
Paimon and
unified
Catalog**

From 25.3 to 26.3 LTS, ClickHouse data lake analytics capabilities have been significantly upgraded:

- **Native Apache Paimon support:** Version 26.3 LTS provides the native paimon table function and Paimon table engine (including cluster-parallel query through paimonClu

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ster). You can perform efficient read-only analytics on Paimon tables stored in OSS, S3, or HDFS, which is ideal for Flink CDC data lake ingestion scenarios.

- **DataLakeCatalog auto-mounting:**

External metastores such as Hive Metastore, AWS Glue, and Alibaba Cloud DLF can be mounted directly as ClickHouse databases. The engine automatically detects Iceberg, Paimon, Delta Lake, and Hudi tables in the catalog without manual table creation.

- **Multi-dimensional data lake query acceleration:**

- **Parquet footer SLRU cache:**
Enabled by

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default
with
use_pa
rquet_m
etadata
_cache
= 1 . I/O
read
requests
can be
reduced
by up to
50% for
repeated
queries
on the
same
Parquet
files.

- **Iceberg metadata async prefetching:** The iceberg_metadata_async_prefetch_period_ms

parameter keeps local metadata fresh in the background, eliminating remote catalog bottlenecks.

- **S3Queue incremental pull:** Ordered mode now uses StartAfter

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semantic
s to
reduce
ListOb
jects
frequenc
y,
lowering
OSS API
costs
and
latency.

Full-text search: Native inverted index GA

The native
inverted index
and full-text
search
capabilities in
2.6.3 LTS are now
production-ready
(GA).

- **Multi-fold
performance
improvement**

: In text
fuzzy
matching
and keyword
search
scenarios,
cold query
performance
improves by
7x to 10x
with the
inverted
index
enabled. In
the GitHub
Events log
query
benchmark,
complex
search
latency

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drops from
193 seconds
to 0.42
seconds.

- **Seamless
SQL
compatibility**

: No external
Elasticsearch
or third-
party search
plugins are
required.
Create an
index by
using
INDEX
idx_text
content
TYPE
inverted
to
automaticall
y accelerate
HAS() ,
LIKE /
ILIKE , and
JSON field
extraction
queries.

- **Low storage
and compute**

costs: Based
on the
column-store
inverted
index
structure of
ClickHouse,
the index
size is much
smaller than
that of
traditional
search
engines,
which
significantly
reduces
storage
costs for log,
trace, and
text

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analytics
scenarios.

Compute and optimizer

Materialized CTE

Version 26.3 LTS introduces the materialized CTE syntax. By using the `MATERIALIZED` keyword, ClickHouse stores CTE results in a temporary in-memory table to avoid redundant computation and improve the execution efficiency of complex reporting and multi-dimensional analysis SQL queries.

Example:



```
SET enable_materialized_cte=1;

WITH top AS (
  SELECT *
  FROM top
  ORDER BY
  LIMIT 1000000
)
SELECT *
FROM top
INNER JOIN
```

Intelligent
JOIN
reordering for

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all JOIN types

The cost-based optimizer (CBO) in 26.3 LTS now supports automatic Build/Probe side swapping for LEFT ANTI JOIN , SEMI JOIN , and FULL JOIN , in addition to standard INNER JOIN and LEFT/RIGHT JOIN . The optimizer automatically builds the hash table from the smaller side based on table statistics, preventing out-of-memory (OOM) errors caused by improper manual SQL ordering.

Storage engine and data types

Sharded Map

For scenarios where the Map type stores user profile tags and dynamic features, version 26.3 LTS introduces a bucketed physical serialization

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layout. Single-key lookups improve by 2x to 49x because data within the Map is hash-bucketed by key.

Example:



```
CREATE T
  id U
  attr
) ENGINE
ORDER BY
SETTINGS
```

Wide table TTL cleanup: Vertical merge

Version 26.3 LTS introduces a vertical merge algorithm for TTL DELETE operations. The merge prioritizes primary key and TTL column filtering without reading irrelevant wide columns, which significantly reduces I/O and memory overhead during background data cleanup.

Adaptive lossless floating-point codec (ALP)

A new lossless floating-point compression codec CODEC(ALP,

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ZSTD) is available. ALP (Adaptive Lossless floating-Point) is optimized for Float32 and Float64 data in industrial monitoring and time-series metrics scenarios, delivering better compression ratios and decompression speed than the classic Gorilla encoding.

Open ecosystem: Cross-engine SQL dialect compatibility (Polyglot)

By integrating the open source Polyglot parsing library, version 26.3 LTS natively supports 31 external database SQL dialects, including Snowflake, BigQuery, PostgreSQL, Spark, Presto, and DuckDB. You only need to configure the following settings:

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SET dial

After that, you can reuse your existing SQL workloads.

Key performance optimizations

Query execution and optimizer

- Query condition cache, expression JIT compilation, and JOIN runtime filters are enabled by default to reduce repeated scans and large table JOIN overhead.
- RIGHT JOIN and FULL JOIN use ConcurrentHashJoin, with up to 2x performance improvement in some scenarios. A new JOIN ordering optimization automatically reorders multi-table

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JOINS based
on statistics.

- Primary key
and partition
key index
pruning is
extended:
primary keys
support
arbitrary
deterministic
expressions
for data
skipping, and
partition
keys
wrapped in
deterministic
function
chains can
still be
pruned.
- ORDER BY
... LIMIT
N queries
can
significantly
reduce
scanned
rows through
skip indexes
and dynamic
threshold
filters.

Storage engine and data skipping

- Parquet
reader v3 is
enabled by
default with
page-level
filter
pushdown
and
PREWHERE
support.
Data lake
reads
automaticall
y adjust

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pipelines
based on
processing
threads,
achieving
approximately 40x
improvement
on multi-core
machines.

- Streaming
skip index
filtering
during reads
is enabled by
default. Text
indexes
support
more
predicate
forms and
can be used
in
PREWHERE.
- In heavy
partition
pruning
scenarios,
SELECT
queries on
tables with
10,000+
parts are up
to 8x faster.
Iceberg
tables
support
PREWHERE
optimization
and async
metadata
prefetching.

Distributed and parallel execution

- Parallel
distributed
INSERT
SELECT is
enabled by
default and
executed

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independently on each shard. Distributed IN subqueries automatically add DISTINCT to reduce cross-shard temporary data transfer.

- Distributed index analysis supports SharedMerge Tree and shared storage scenarios. Parallel replicas support lazy materialization with improved load distribution to reduce long-tail latency.

Functions and indexes

- LIKE and regular expressions are automatically rewritten to more efficient implementations and enabled by default. StringZilla accelerates case-sensitive

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string
search, and
SIMD
dynamic
dispatch
accelerates
logical
functions
and boolean
column
conversion.

Memory, observability, and networking

- Mark, uncompress ed, and page caches use independent jemalloc arenas to reduce fragmentation. jemalloc dirty pages are cleaned in independent threads. System log tables add minmax and bloom_filter indexes to accelerate troubleshooting.
- Block serialization and compression in distributed queries can be offloaded to pipeline threads to improve large-volume data transfer efficiency.

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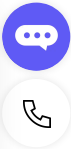


Performance
benchmark:
ClickBench
comparison

Based on the official ClickBench version benchmark (Cold Queries perspective), version 26.3 LTS achieves continuous performance improvement compared to the previous LTS version 25.3:

Metric	Improvement of 26.3 LTS over 25.3 LTS
Cold query latency	8% to 15% reduction in total time
Data access	2x to 5x speed

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la k e a n d st or a g e s c a n	up for I/O- intensi ve querie s and OSS extern al table scans
W id e ta bl e T T L c o m p a ct io n	Peak memor y usage reduce d by more than 40%
T e xt a n d lo g s e ar c h	Compl ex search latenc y reduce d from 193 secon ds to 0.42 secon ds with inverte d index
Hi g	2x to 49x

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h-	improv
di	ement
m	in
e	single-
n	key
si	lookup
o	with
n	sharde
al	d Map
M	
a	
p	
p	
oi	
nt	
lo	
o	
k	
u	
p	

Utility features

Version 26.3 LTS also includes a set of features that improve the development and O&M experience:

- **Natural sort** (**naturalSortKey**): Sorts strings containing numbers in human-intuitive order. For example, `file2.txt` is correctly sorted before `file10.txt`.
- **Tree-form EXPLAIN**
output: Run EXPLAIN

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pretty=1,
compact=1
to obtain a
well-
structured,
visual
execution
plan tree.

- **Unavailable
shard fault
tolerance:**

The
max_skip_
unavailab
le_shards
_num and
max_skip_
unavailab
le_shards
_ratio
parameters
provide
bounded
control over
the ratio of
fault nodes
that can be
skipped
during large
table queries
in a cluster.

Upgrade recommendations

Pre-upgrade preparation

1. **Clone or test
environment
verification:**
Deploy the
new version
in a cloned
or test
environment
first to
evaluate the
impact of
version
changes on

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existing
configurations
and
queries.

2. **Check
dependency
compatibility**
: Confirm the
compatibility
of client
drivers and
third-party
tools with
the new
version.

Important notes

- **Experimental
feature
validation:**
Materialized
CTE and
Polyglot
dialects are
marked as
Experimental
in this
version. We
recommend
that you
validate
these
features in a
test cluster
before using
them in
production.
- **Performance
regression
testing:** After
the upgrade,
compare the
execution
time of key
queries to
ensure that
optimizations
such as
Parquet
cache,
inverted
index, and

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JOIN
reordering
take effect.

- **Monitoring and alerting adjustment:**
Adjust
monitoring
thresholds
based on
new metrics
and
changes.

Summary

ClickHouse 26.3
LTS is a release
of both breadth
and depth:

- At the **data lake and open ecosystem**
level, native
Paimon
support and
compatibility
with 31 SQL
dialects
bridge
lakehouse
integration
and cross-
engine
reuse.
- At the **query and analytics**
level, GA
inverted
indexes,
materialized
CTEs, JOIN
reordering,
and sharded
maps
significantly
expand
ClickHouse
capabilities
in log search,

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complex
analysis, and
high-
dimensional
point
lookups.

- At the **stability and cost** level, TTL vertical merge and ALP Codec reduce operational complexity across data cleanup and storage compression.

The Alibaba Cloud ClickHouse team has completed cloud adaptation and full compatibility verification for the community 26.3 LTS release. You can quickly create a 26.3 LTS instance or upgrade your existing instance to the latest version in the Alibaba Cloud ClickHouse console to experience more efficient and stable real-time analytics.

Next:
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